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Lab session 1

S1.1

- Measure the width of the resist at different Z Positions: 10 μm , 50 μm and 80 μm and show their values.

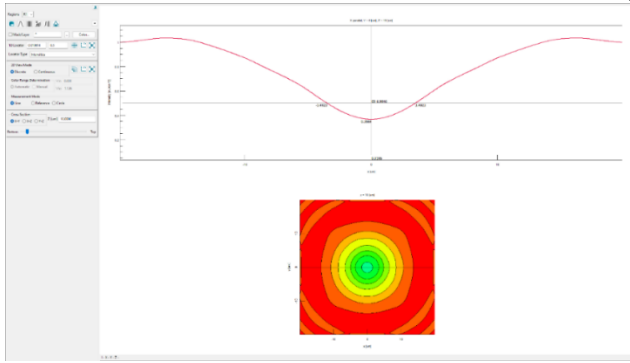


Figure 1 Width of resist at Z=10 μm

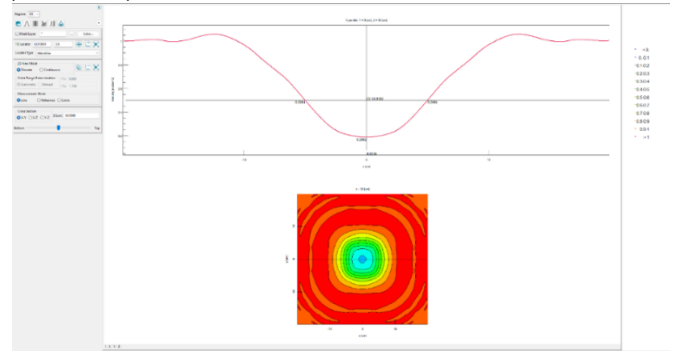


Figure 2 Width of resist at Z=50 μm

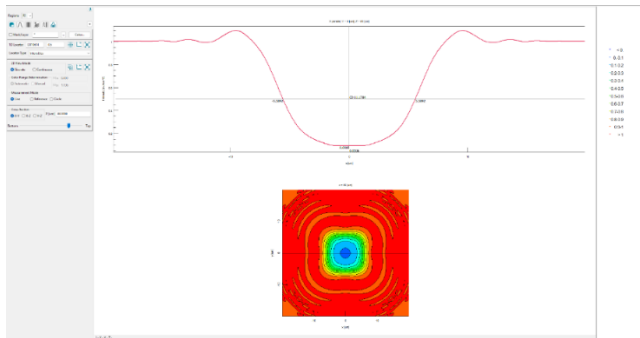


Figure 3 Width of resist at Z=80 μm

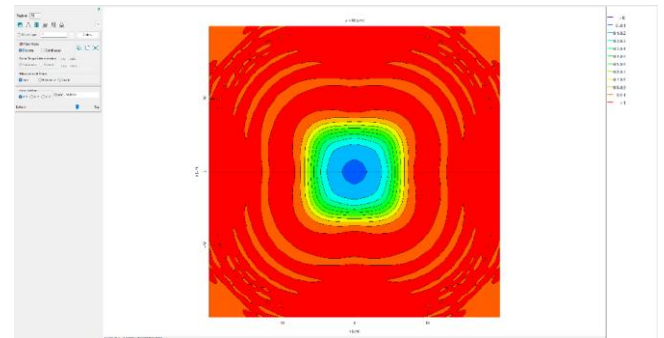


Figure 4 2D Contour fill plot for Z=80 μm

Table 1 Width of resist at varying Z values

Z(μm)	Width (μm)
10	6.9846 (-3.4923 to 3.4923)
50	10.0162 (-5.0081 to 5.0081)
80	11.1784 (-5.5892 to 5.5892)

Observation:

We can see that Table 1 shows that the width of the resist is also increasing with the increase in the value of Z. This means that the beam is not focused at these depths, but it is divergent. This occurs when the focal plane is either nearer to the surface or even above the resist completely. Thus, the resist will be tapered in the sidewalls.

It also represents a rather low DOF (Depth of Focus) or a high NA (Numerical Aperture) system in which the beam diffuses quickly out of the focal point.

For 2D contour fill plot of Z = 80 μm :

The wider the beam is disseminated across the lower to higher Z values, the wider the intensity profile and the lower the minimum intensity at the centre are, as the dark colour in the centre of the 2D contour in Figure 4 indicates. That is why the 2D contour at larger Z values has more diffraction rings, as the light is interfering more as it spreads [1].

S1.2

- Measure the width of the resist at different Z Positions: 10 μ m, 50 μ m and 80 μ m and compare their values with the dose.

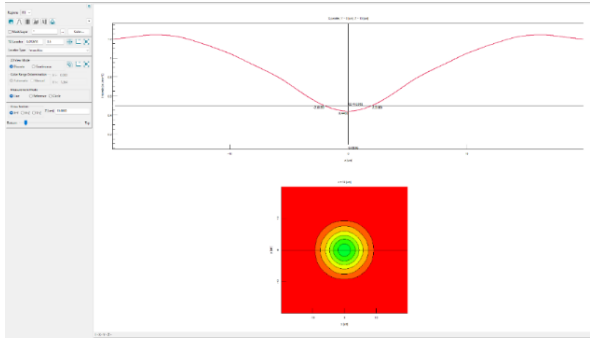


Figure 5 Exposure 1.2mJ/cm² at Z=10 μ m

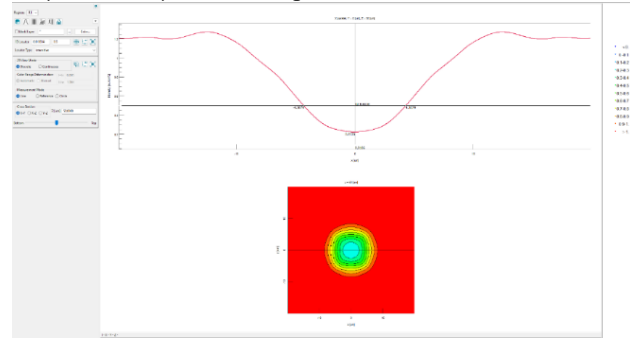


Figure 6 Exposure 1.2mJ/cm² at Z=50 μ m

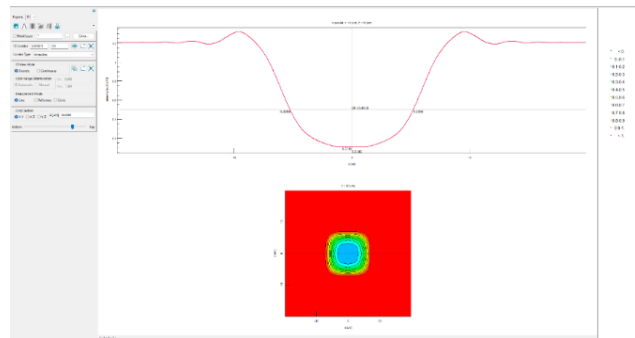


Figure 7 Exposure 1.2mJ/cm² at Z=80 μ m

Table 2 Width of the resist at an Exposure of 1.2mJ/cm²

Z(μ m)	Width (μ m)
10	4.036 (-2.0180 to 2.0180)
50	8.6158 (-4.3079 to 4.3079)
80	10.4516 (-5.2258 to 5.2258)

Observation:

- At higher exposure dose, the intensity is more aggressively decreasing below the 0.5 line of threshold, leading to a substantially narrower width of the resist at all depths.
- We observe that at Z=10 the shrinkage is almost equal to 3 μ m, whereas at Z=80 the shrinkage is almost equal to 0.7 μ m. This implies that the sidewall angle is becoming much steeper and the structure itself is becoming more pyramidal as the top of the resist is much more dose sensitive than the bottom.
- The lowest intensity at Z=80 has also decreased to less than 0.2 compared to the earlier 0.25, which is reflected in a cleaner development of resist. The dark area becomes clearer now, enhancing the level of resolution and minimizing the probability of a leftover resist in the middle of the feature.
- Make the comparison of the x-z cross-section (filled contour) for both doses.

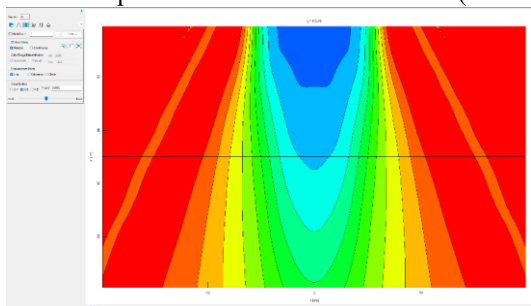


Figure 8 X-Z Cross-section Exposure dose 1.0mJ/cm²

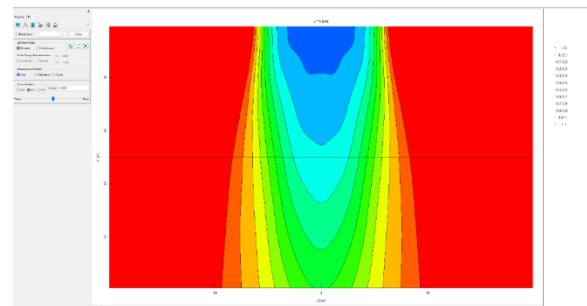


Figure 9 X-Z Cross-section Exposure dose 1.2mJ/cm²

Comparison:

By increasing the dose from 1.0 mJ/cm² to 1.2 mJ/cm², we are pushing the energy deeper and wider into the material.

Exposure dose 1.0 mJ/cm ²	Exposure dose 1.2 mJ/cm ²
The most obvious difference is the dark blue core, which is relatively small and concentrated near the top surface, therefore, a wider resist remains.	The dark blue core expands significantly, both in width and depth, which indicates that a much larger volume of the resist is receiving a high enough dose to clear effectively during development.
The boundary of the green area is more relaxed, which means that it has a shallower slope.	The green region is noticeably wider, which means a higher contrast at depth.
The sidewalls have a more pronounced curve as they go deeper, therefore, it has a more softer and curved shape.	The orange to yellow to green zones appear stretched. The profile becomes more linear and stiff. While the feature is narrower overall, the change in width from top to bottom is more aggressive, which means the side is sharper and tapered.
Diffraction fringes are mostly red (> 0.9 intensity), meaning they are away from the development threshold. Therefore, it can be underdeveloped.	The diagonal wings are becoming more yellow and orange, which means the exposure is rising. It has a risk of sidelobe interference.

S1.3

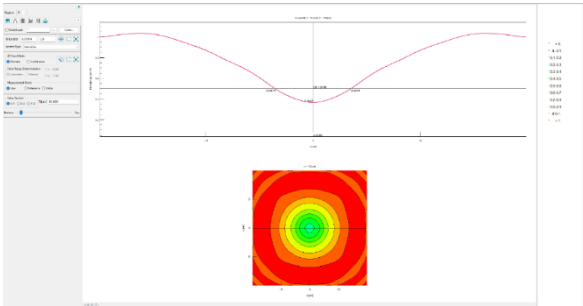


Figure 10 Width of resist for Multiple wavelengths at Z=10 μm

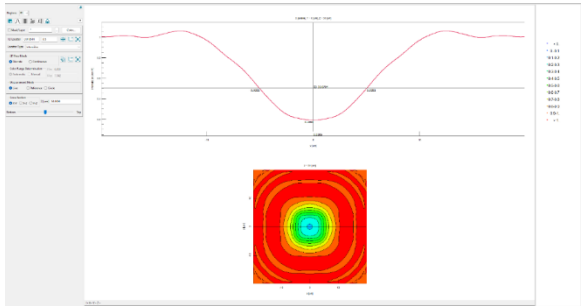


Figure 11 Width of resist for Multiple wavelengths at Z=50 μm

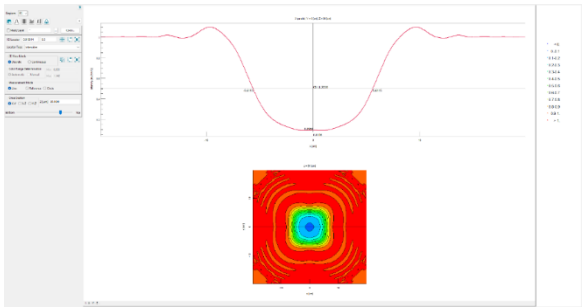


Figure 12 Width of resist for Multiple wavelengths at Z=80 μm

Table 3 Width of the resist for multiple wavelengths

Z(μm)	Width (μm)
10	7.0748 (-3.5374 to 3.5374)
50	10.0704 (-5.0352 to 5.0352)
80	11.223 (-5.6115 to 5.6115)

Observation:

The multicoloured light illumination produces a slightly broader resist feature at all depths. This is due to the fact that in multiple-wavelength lighting, there exists a range of wavelengths with a slight variation in the diffraction angle. This slightly blurs the edges of the exposure and spreads the total energy distribution.

The curves are far smoother since the peaks of interference of the different wavelengths occur at different space positions, effectively cancelling out the sharpest variations of one another. Its overall spread is quite comparable to the single wavelength case, and this indicates that although multiple wavelengths tend to smooth out the local interference, they do not significantly alter the overall beam divergence of the system.

- Make the comparison of the x-z cross-section (filled contour) for both single and multiple wavelengths.

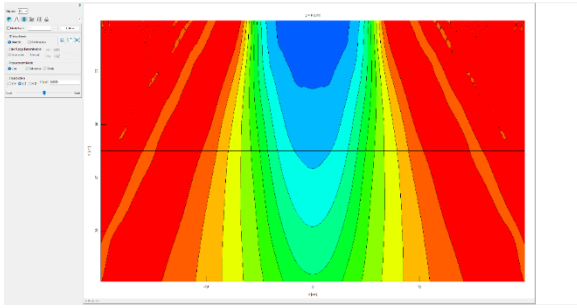


Figure 13 Single wavelength X-Z cross-section (Filled contour)

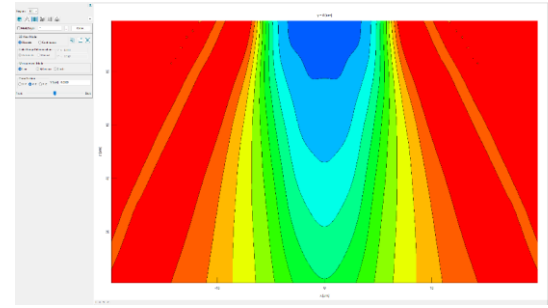


Figure 14 Multiple wavelengths X-Z cross-section (Filled contour)

Comparison:

Single wavelength	Multiple wavelength
We can observe sharp speckles and jagged edges in the diagonal orange and red bands, which are high-contrast interference patterns.	The sidelobes are significantly smoother and more blurry. Because it produces interference peaks at slightly different spatial positions, they average each other out. This results in a much cleaner and more continuous gradient of energy.
The cyan region appears slightly narrower at the edges.	The cyan region appears slightly broader at the edges because different wavelengths focus at slightly different depths, making the beam naturally softened.
Less reliable	It gives a more uniform clearing dose at the surface, which can lead to better process reliability.

Multiple wavelength illumination acts as a spatial filter, cleaning up the noisy interference seen in Figure 13. By doing so, we gain a more stable and smoother resist profile that is less sensitive to wavelength-specific artefacts.

S1.4

- Change the collimation to 0° angle and show 1D and 2D contour intensity image with Z position variations.

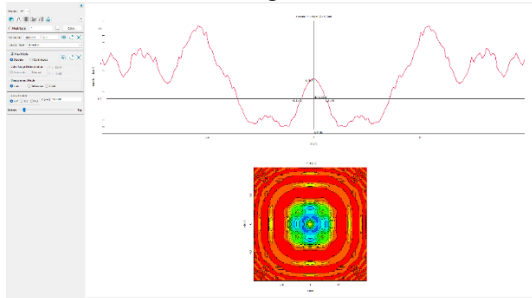


Figure 15 0° Collimation angle Z=10 μm

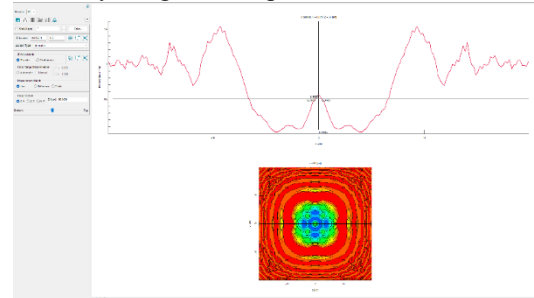


Figure 16 0° Collimation angle Z=50 μm

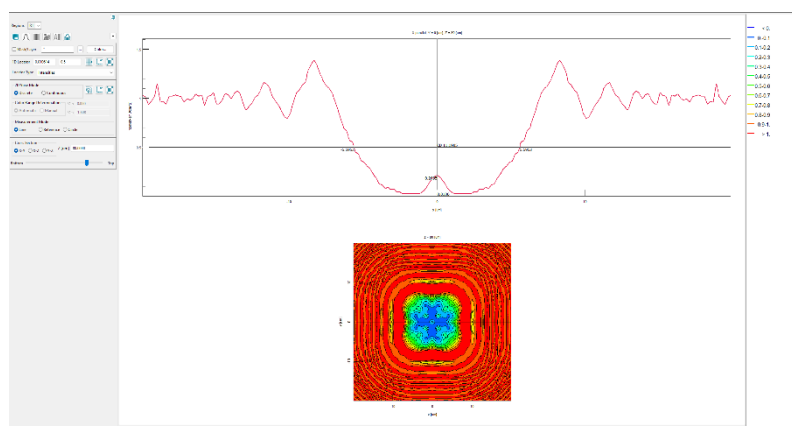


Figure 17 0° Collimation angle Z=80 μm

Observation:

In lithography, a 0° collimation angle typically maximizes resolution and DOF but makes the system extremely sensitive to diffraction and interference. By comparing the 1D intensity image:

At 10 μm : The interference ripples are very high and sharp, which suggests high noise.

At 50 μm : The central spike remains, but the width of the primary dark well is stable as compared to 10 μm .

At 80 μm : The centre spike almost vanishes, and the profile becomes smoother.

- Compare light intensity at 0° and 4° angle and show their 1D and 2D intensity at $z=50\mu\text{m}$ and discuss the influence of collimation angle.

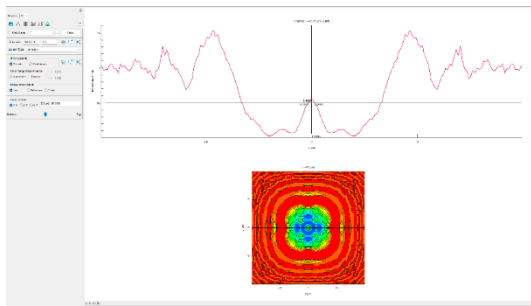


Figure 18 0° Collimation angle $Z=50 \mu\text{m}$

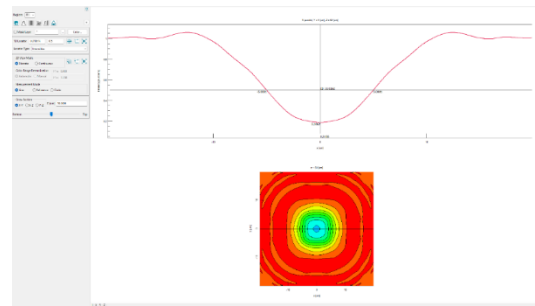


Figure 19 4° Collimation angle $Z=50 \mu\text{m}$

Comparison:

0° collimation	4° collimation
Light rays are perfectly parallel, which results in severe Fresnel diffraction. We can see sharp, high-frequency ripples in the background and a distinct central intensity peak inside the feature.	The angular spread of light smears these interference patterns. The ripples in the background are virtually eliminated, and the central intensity peak is gone, resulting in a smooth, monotonic well.
The transition from dark center to the bright background is much steeper, which allows more vertical sidewalls and better resolution.	The edges are slightly blurred, which results in a more tapered sidewall in the developed resist.
The center of the feature is dangerously close to the threshold, which might unintentionally leave a defect.	The center is well below the threshold, providing a much larger process window.
The ripples in the pattern are much more pronounced, which might lead to a line edge roughness.	The pattern is much more uniform and circular. The energy distribution is clean, which translates to smoother resist surfaces.

S1.5

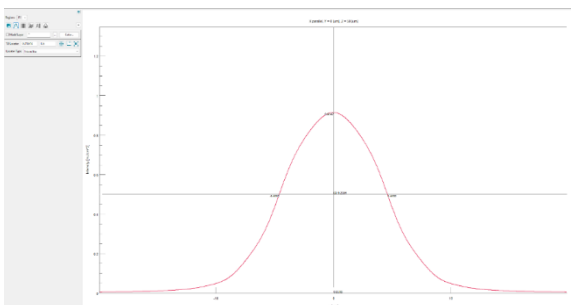


Figure 20 1D Dark-field intensity at $Z=50 \mu\text{m}$

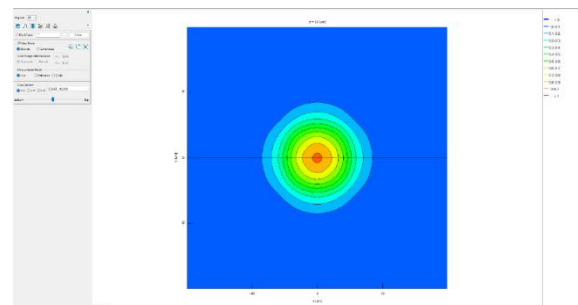


Figure 21 2D Dark-field intensity at $Z=50 \mu\text{m}$

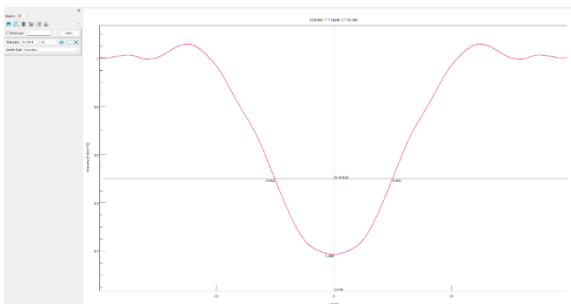


Figure 22 1D Bright-field intensity at $Z=50 \mu\text{m}$

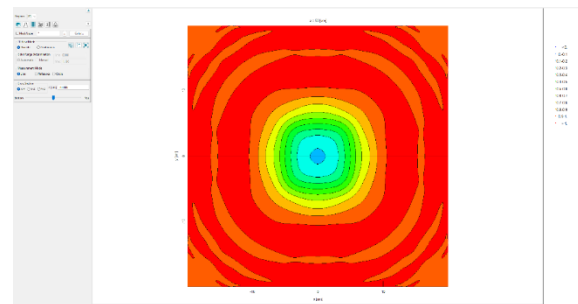


Figure 23 2D Bright-field intensity at $Z=50 \mu\text{m}$

Comparison:

Dark-field	Bright-field
The mask is opaque everywhere except for a small opening in the center, which produces a peak of intensity which reaches approximately 0.9 and drops to 0 at the edges [2].	The mask is transparent everywhere except for a small opaque feature at the center, because of which the intensity drops to 0 in the center and rises to 1 at the edges [2].
There is almost zero light intensity at the outer regions and high in the center.	There is high intensity at the outer regions and almost zero in the center.

S1. 6

Table 4 Comparison of the accuracy of the width of the resist at $Z=50\text{ }\mu\text{m}$ with varying grid sizes

Grid X/Y	Width of the resist (μm)
2.0	9.7046 (-4.8523 to 4.8523)
1.0	9.8586 (-4.9293 to 4.9293)
0.8	9.9414 (-4.9707 to 4.9707)
0.6	9.9796 (-4.9898 to 4.9898)
0.4	10.0082 (-5.0041 to 5.0041)
0.2	10.0162 (-5.0081 to 5.0081)
0.1	10.038 (-5.019 to 5.019)

Observation:

- The coarse grids (such as 2.0 or 1.0) are more likely to smooth out the intensity gradients. Since the grid points are not close to each other, the simulation loses the precise point at which the intensity reaches the 0.5 mark and therefore underestimates the width of the resist.
- It is true that the finer the grid size, the more time the computation takes and the more memory consumed. The resist width of grid 0.1 is not significantly different from the resist width of grid 0.2, and this implies that an additional reduction in grid size will negatively impact the actual value of the resist width.

S1. 8

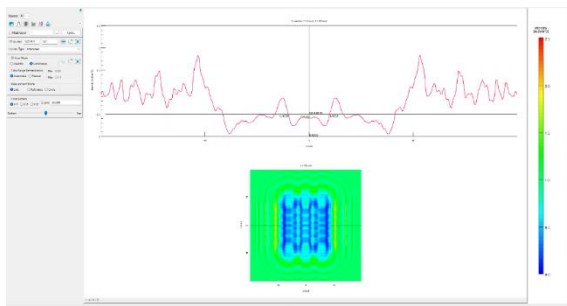


Figure 24 Initial bright-field representation

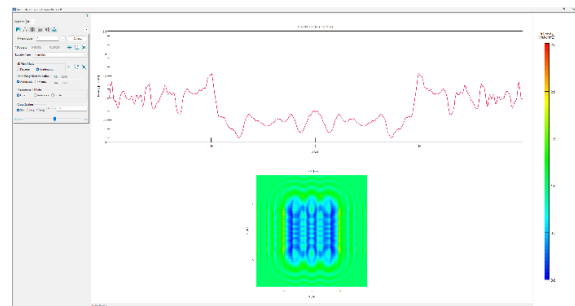


Figure 25 Inversion of the graph

Position of Z for inversion to occur = $57\text{ }\mu\text{m}$

Observation:

When we observe task 5, it is evident that the dark-field mask produces an intensity peak which corresponds to the openings in the center of the mask, similarly, the bright-field mask produces a dip in intensity at the center which corresponds to the opaque regions in the mask. The same reactions can be observed in Figure 24 and Figure 25.

Also, when we observe the 2D plot, it is evident that the bright-field mask has higher intensity in the outer regions than in the central part. Similarly, the dark-field representation has higher intensity at the center than at the outer regions, which represents absorption in the outer regions [2][3].

S1. 9

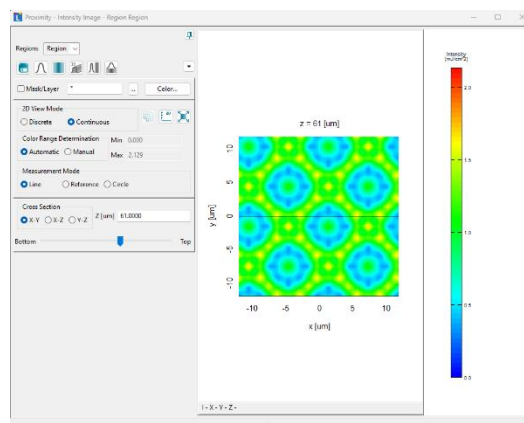


Figure 26 Replicated aerial image at $Z=61$ and Collimation angle=0.85

Observation:

- The Z-position is one of the most critical parameters for the periodic mask because, as we vary its value, the intensity distribution of the mask changes. The exact image of the mask is replicated at a Z distance of 61 μm , but as we move away from this Z value, the pattern becomes blurry.
- Similarly, for a low collimation angle, the pattern is much sharper and of high contrast, but as we increase the angle i.e., 0.85, the pattern averages out the peaks and becomes smoother as shown in Figure 26.

S1 10.

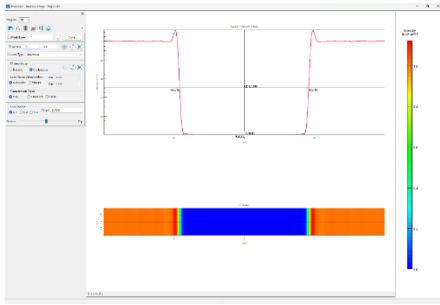


Figure 27 Tuning for 3 μm

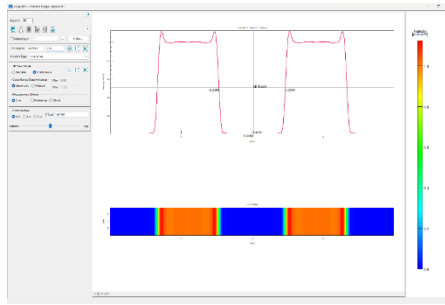


Figure 28 Tuning for 1.5 μm

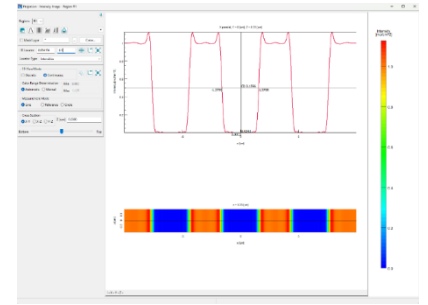


Figure 29 Tuning for 1 μm

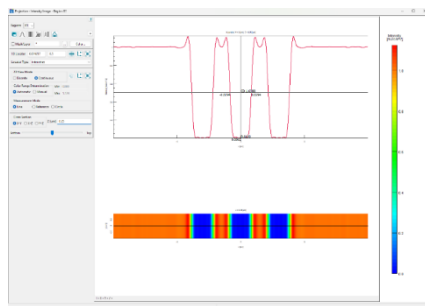


Figure 30 Tuning for 500 nm

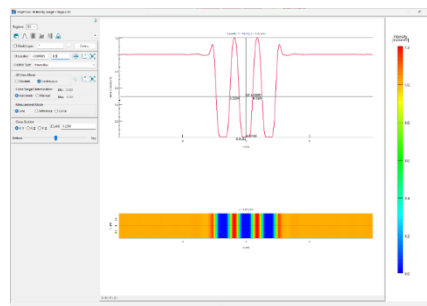


Figure 31 Tuning for 300 nm

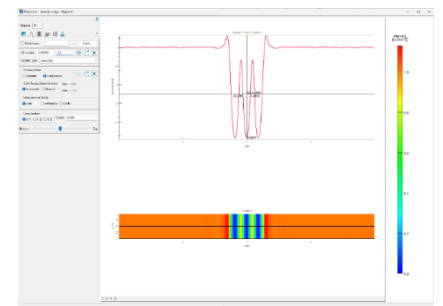


Figure 32 Tuning for 150 nm

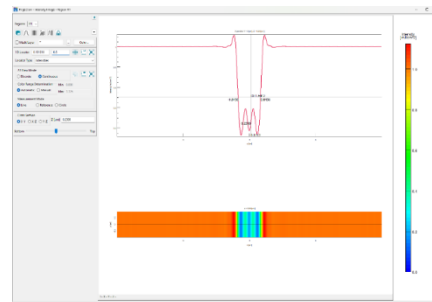


Figure 33 Tuning for 100 nm

Table 5 Tuning the width of the middle line

Z	Width (μm)
3 μm	9.1596 (-4.5798 to 4.5798)
1.5 μm	4.6598 (-2.3299 to 2.3299)
1 μm	3.1596 (-1.5798 to 1.5798)
500 nm	1.6588 (-0.8294 to 0.8294)
300 nm	1.0588 (-0.5294 to 0.5294)
150 nm	0.5698 (-0.2849 to 0.2849)
100 nm	1.6912 (-0.8456 to 0.8456)

Observation:

$$R = 0.6 * \frac{\lambda}{\text{NA}}$$

Based on the given parameters, the calculated resolution is 445.3 nm.

The closest distance between two features that the system is capable of perceiving as separate is known as resolution. Since wavelength and NA of the system are fixed resolution is a constant over most of the range of variation of target width, but it fails to compute the correct resolution at 100 nm [4].

As it is evident from Table 5 that as the value of Z decreases, the width of the resist also decreases, but at 100 nm, the width suddenly increases. This occurs as the features become smaller and smaller, the separation between them, the light emitted through one aperture interferes with that emitted through the other aperture, i.e. sidelobe interference. Due to this, the dark area between them gets sufficient leaked light that the intensity never falls below the 0.5 level. This interference in the lines forms one blurry spot due to the fact that the system failed to resolve the gap between the lines, and therefore, failed to measure the appropriate resolution [3].

S1 11.

Table 6 Tuning the width for NA = 0.25

Z	Width (μm)
3 μm	9.30 (-4.65 to 4.65)
1.5 μm	4.818 (-2.4090 to 2.4090)
1 μm	3.318 (-1.6590 to 1.6590)
500 nm	1.813 (-0.9065 to 0.9065)
300 nm	1.1274 (-0.5637 to 0.5637)
150 nm	2.5866 (-1.2933 to 1.2933)
100 nm	1.674 (-0.8370 to 0.8370)

Observation:

For NA=0.25

Updated resolution = 890.6 nm

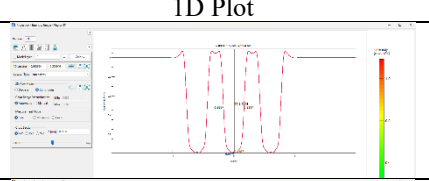
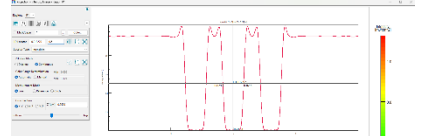
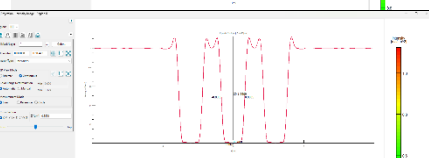
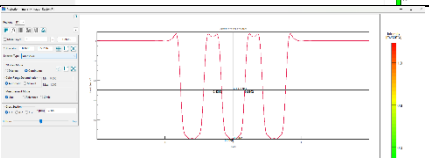
Reducing the NA from 0.5 to 0.25 significantly degrades the resolution of the system. A lower NA means the lens collects fewer diffracted orders, which blurs the aerial image and increases the minimum resolvable feature size.

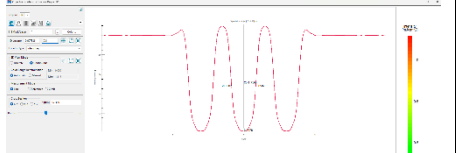
- The initial measurement is that the width of the mask narrows as we tune it, but the system no longer works, i.e. at 300 nm, which is way too short of the calculated Rayleigh limit of 890.6 nm.
- The width rises to 2.5866 μm at 150 nm, meaning that the lines have become a blurry blob due to the optics being unable to separate them.
- At 100 nm, it is basically a random diffraction noise; hence, the width is narrowed.
- Small features bend light at steep angles. A lens of NA = 0.25 would not have a large enough aperture to see these steep angles and would essentially serve as an LPF that would completely remove all the sharp edges of the mask.
- This is also very sensitive to the slightest dose or the 0.5 threshold, and results in huge biases in measurement, as seen in Table 6 [2][4].

S1 12.

For NA=0.5

Table 7 Defocus variation for NA = 0.5

Defocus	Width (μm)	1D Plot
-1	1.6794 (-0.8397 to 0.8397)	
-0.5	1.6597 (-0.8299 to 0.8299)	
0	1.6609 (-0.8305 to 0.8305)	
0.5	1.6783 (-0.8392 to 0.8392)	

1	1.7164 (-0.8582 to 0.8582)	
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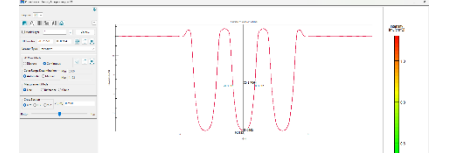
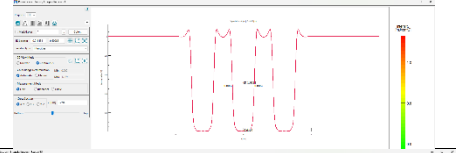
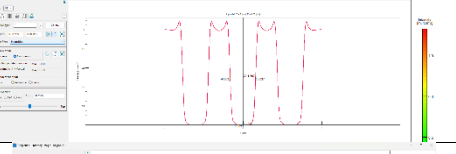
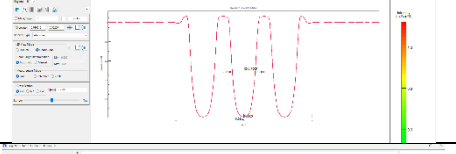
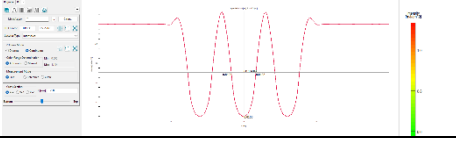
Observation:

The distance between the resist surface and the desired focal plane of the lens is termed defocus. As Table 7 shows, the minimum width is achieved at -0.5 and not at 0, implying that the optimal focal plane in this particular air stack and resist combination is a little way into the material.

- With increased defocus, the shoulders of the bright areas were rounded, and the slope of the intensity curve at a 0.5 threshold was made shallower. A shallow slope is a hazardous aspect of manufacturing since this implies that a small shift in exposure dose will result in a colossal shift in the corresponding line width.
- This will cause the difference between the light and dark regions to decrease [4].
- Looking at the 1D plot for defocus = 1, the interference ripples at the edges of the pattern start to flatten out. This loss of high-frequency detail is the first sign that we are reaching the limit of DOF.

For NA=0.8

Table 8 Defocus variation for NA = 0.8

Defocus	Width (μm)	1D Plot
-1	1.7064 (-0.8532 to 0.8532)	
-0.5	1.6123 (-0.8062 to 0.8062)	
0	1.6073 (-0.8037 to 0.8037)	
0.5	1.7005 (-0.8503 to 0.8503)	
1	1.6844 (-0.8422 to 0.8422)	

Observation:

Increasing the NA to 0.8 significantly improves the resolution, but at the cost of a narrower DOF.

- The slope of the intensity curve at 0.5 threshold is far more vertical. It means a greater Image Log Slope (ILS), providing a significantly greater control over features and steeper sidewalls.
- The system is collecting more diffracted light, leading to a much better image contrast.
- We get the minimum width at defocus = 0, which means that the system is now centered correctly, and the aerial image is cleaner and closer to the ideal mask shape at best focus.
- DOF is negatively proportional to $(NA)^2$, and the light rays enter at much steeper angles. Due to this, a high NA system demands far greater precision of substrate levelling and thickness control [4][5].

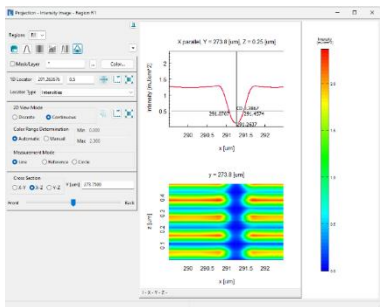


Figure 34 Single wavelength XZ

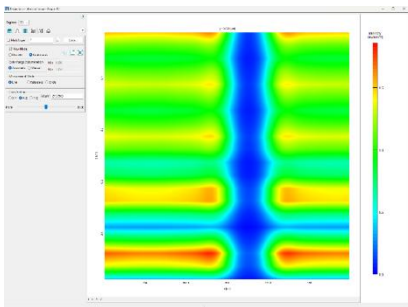


Figure 35 Multiple wavelength XZ

Observation:

Single wavelength XZ	Multiple wavelength XZ
The interference fringes are very sharp and periodic. The contrast between the horizontal dark and bright layers is high because it maintains perfect phase coherence throughout the depth of the resist. The horizontal lines are sharpened up as we move into the deeper material.	The horizontal bands also remain visible, though they are blurred in comparison with Figure 34. This is due to the fact that various wavelengths form standing wave peaks at a slightly different Z position. The maximum of one wavelength coincides with the minimum of another, and averages out the total intensity distribution. This is a good lithography effect since the resist sidewalls no longer have a staircase profile after development. In Figure 35, the horizontal bands are greatly obscured as we proceed farther into the material.
We can see sharp speckles in the red/orange zones as we go deeper; these are high-order diffraction fringes. Because the light is monochromatic, these constructive interference points are very localised and hard to detect.	The diffraction fringes are much smoother because the energy spreads over a wider area, turning the sharp interference spots into a continuous and soft gradient.
It indicates that the single-wavelength source provides an inconsistent dose through the thickness of the resist.	The bottom intensity is smoother and closer to the average yellow/green background, which implies that the dose is much more uniform across the thickness of the resist, thereby enhancing the process window.

Using multiple wavelengths acts as a natural noise filter for the optical system. While it slightly reduces the peak contrast at the edges, it significantly improves the physical quality of the resist profile by eliminating standing wave artefacts [5][4].

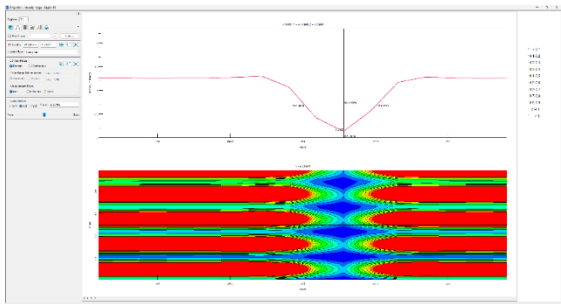


Figure 36 Single-wavelength Si substrate

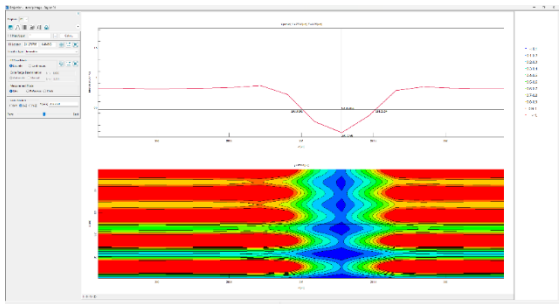


Figure 37 Multiple wavelength Si substrate

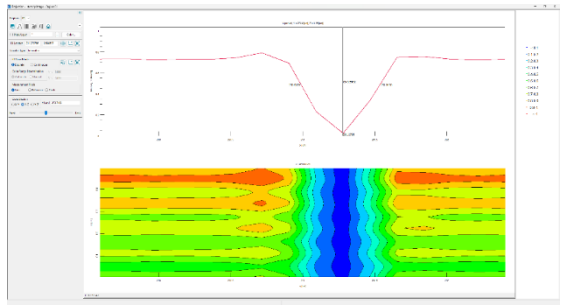


Figure 38 Single-wavelength SiO₂ substrate

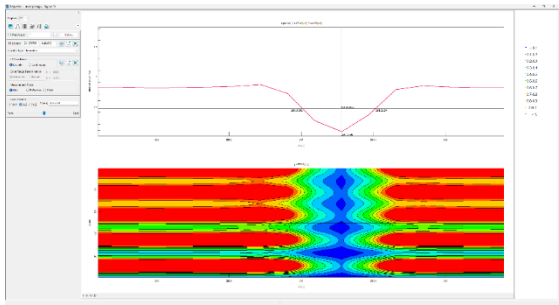


Figure 39 Multiple-wavelength SiO₂ substrate

Observation:

The substrate material determines the amount of light reflected back into the resist, which dictates the strength of the interference [2].

Substrate comparison:

Si	SiO ₂
It has a very high refractive index contrast with the resist, causing strong reflections. We can notice the hard horizontal bands in Figures 36 and 37. The contrast between the red (constructive) and blue (destructive) zones is extreme.	It acts as a partial buffer because its refractive index is closer to that of the photoresist. Figures 38 and 39 have much softer horizontal transitions. The intensity distribution is much more uniform throughout the depth of the resist.
These strong standing waves create deep scallops on the resist sidewalls. Without a BARC, it is very difficult to get a smooth vertical wall on silicon.	In Figure 38, the overall background is more yellow/green, indicating a more balanced energy distribution.

Wavelength comparison:

Single wavelength	Multiple wavelength
In Figures 36 and 38, it is observed that the single wavelength produces perfectly periodic interference. The blue spots are localized and sharp, but the interference is very sensitive in this case.	Since each wavelength has its own interference period, the peak of one wavelength fills in the valleys of another, which gives a smoother resist profile. In Figures 37 and 39, the interference is almost entirely suppressed, resulting in a very smooth vertical shadow in the center.

The best aerial image for resist development is Figure 39 (Multiple wavelengths on SiO₂). It combines the lower reflectivity of the oxide substrate with the spectral averaging of multiple wavelengths to produce the most uniform intensity profile.

Figure 36 (Single wavelength on Si) represents the most challenging process condition, where standing wave artefacts are at their absolute maximum, leading to poor sidewall quality and high CD sensitivity [5].

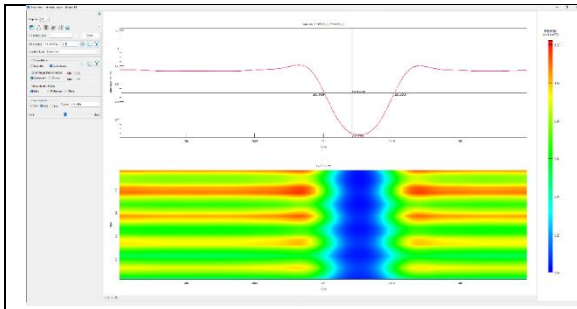


Figure 40 Thickness = 100 nm

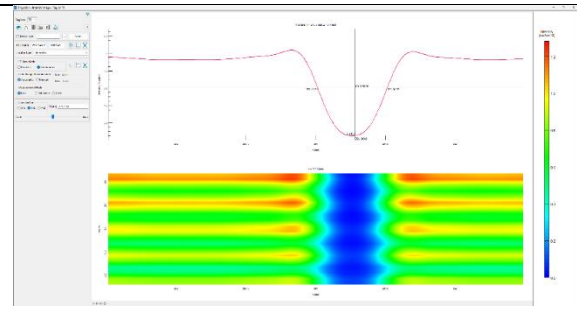


Figure 41 Thickness = 150 nm

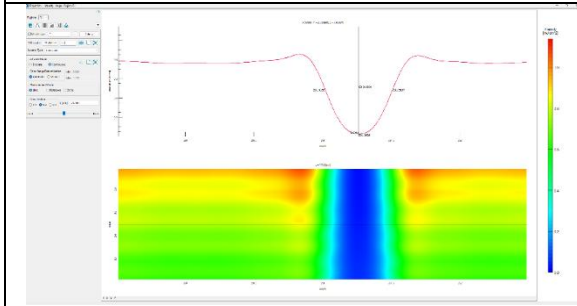


Figure 42 Thickness = 200 nm

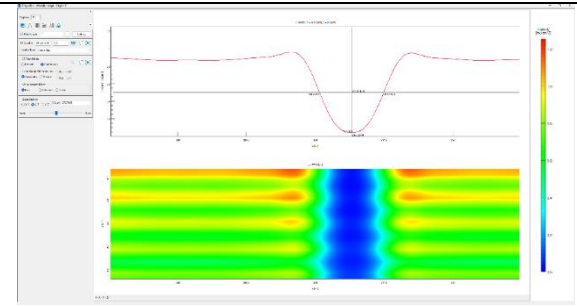


Figure 43 Thickness = 250 nm

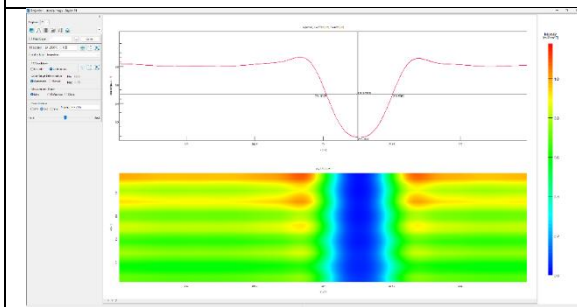


Figure 44 Thickness = 300 nm

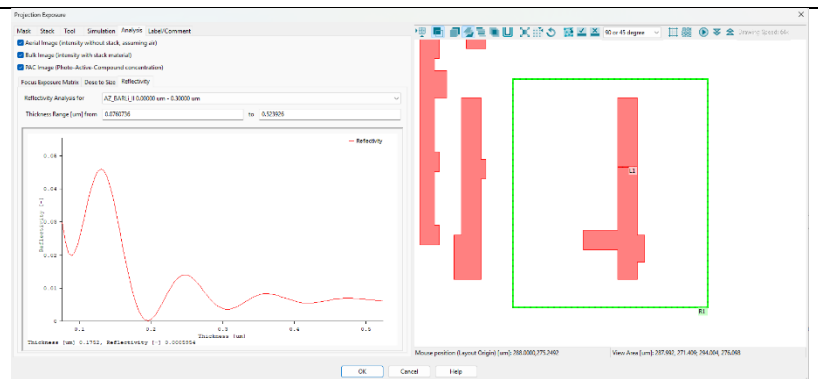


Figure 45 Analysis of reflectivity at 300 μm

Observation:

At a thickness of 100 nm: The resist shows sharp horizontal bands, meaning that a significant fraction of the light is still bouncing off the substrate, leading to the formation of nodes and antinodes. This would result in scalloped resist sidewalls.

At BARC thickness of 150 nm -200 nm: There is blurring and loss of contrast in the horizontal bands. The intensity distribution is made more continuous in the vertical.

In the case of BARC thickness of 250 nm -300 nm: The interference fringes are nearly completely suppressed. The vertical shadow of the mask is also smooth and uniform, and this proves that the BARC is actually absorbing the light before it can reach the substrate and go back.

Optimal BARC thickness:

According to Figure 45, one can conclude that the reflectivity graph exhibits multiple local minima, though 100 nm (first valley) is typically utilized in thin-film processes, there is nevertheless certain interference as observed in Figure 40.

The graph shows that there must be an optimal thickness that occurs around 185 nm to 195 nm with a deep minimum in reflectivity of around 0.0006. This offers a high absorption level with an etchable film thickness.

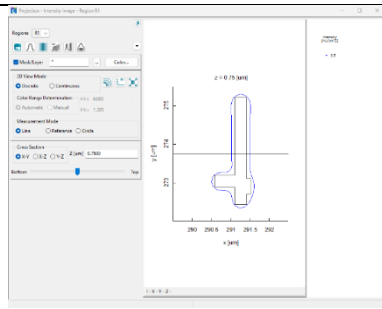


Figure 46 Contour line intensity for dose = 1

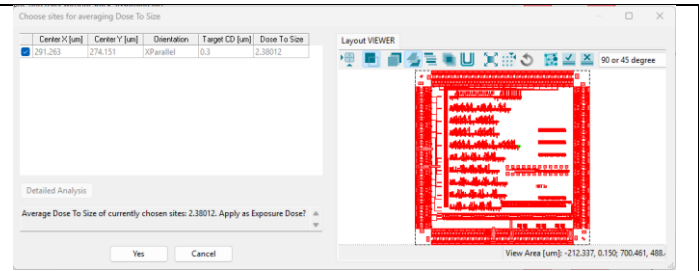


Figure 47 Dosage for top

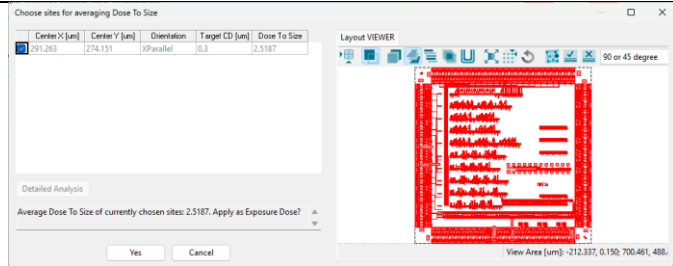


Figure 48 Dosage for centre

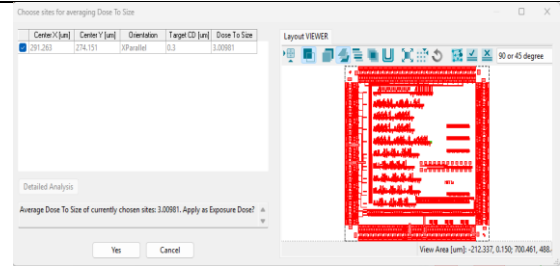


Figure 49 Dosage for bottom

Table 9 Dose to size analysis for Exposure dose

Height	Exposure dose
Top	2.38012
Center	2.5187
Bottom	3.00981

Observation:

In lithography, dose to size analysis is a process that quantifies the exposure energy necessary to form a given resist feature to the desired CD at various depths in the photoresist.

When passing through the resist material, light is scattered. As a result, the intensity of light at the bottom is much less than the intensity of light at the top. The total exposure dose has to be higher in order to counter this loss of energy and attain the same base target size of 0.3 μm .

Top-to-bottom substrate reflections often intensify this large jump in exposure dose. When the bottom of the resist is close to a destructive node, a much larger dose is needed to remove the resist to the desired size and move the intensity threshold to the desired spatial location. When we take the Top dose, the bottom of the resist will be under-exposed, and the sidewalls will be tapered with the bottom of the resist being wider than the top. Likewise, with a Bottom choice, the Bottom will be over-exposed, so that the top may tend to run into top-loss, in which the features are smaller than desired on the surface [5].

S2. 17

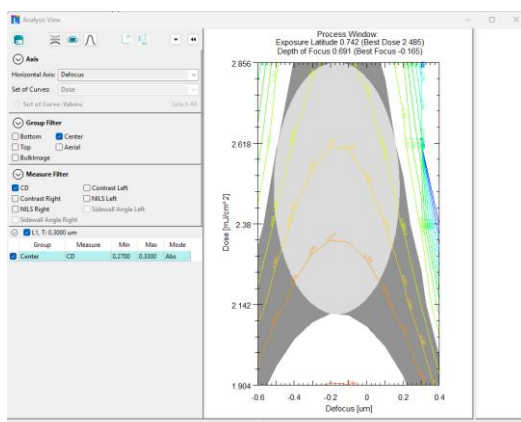


Figure 50 Focus exposure matrix

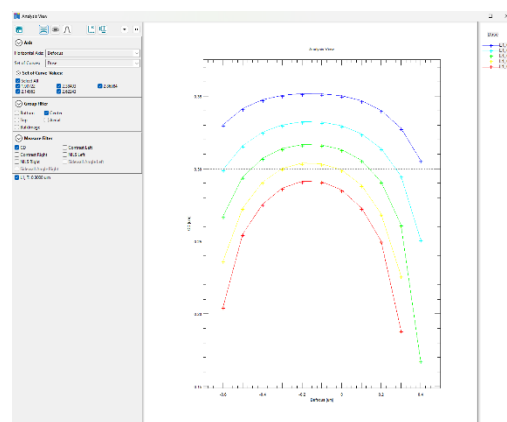


Figure 51 Analysis view

Observation:

The focus exposure matrix is a tool used in lithography to determine the stability of a process. This plot is often called a Process Window, which illustrates how the CD of the feature changes as we vary both the Focus and Exposure dose [5].

From the graph, the following parameter values can be extracted:

Parameter	Value	Definition
Exposure latitude (EL)	0.742	The percentage range of dose variation allowed while staying within CD specifications.
DOF	0.691 μm	The total vertical range of focus movement allowed at the best dose.
Best dose	2.485 mJ/cm^2	The energy required to hit exactly 0.3 μm at the best focus.
Best focus	-0.165 μm	The optimal focal plane for this resist stack.

The light gray elliptical area in the centre represents the Process Window. Any combination of dose and focus within this ellipse will result in a resist feature that meets the 270-330 nm specification.

A DOF of 0.691 μm is quite healthy for an i-line process. This means that the wafer can have surface topography errors of up to 345 nm in either direction without causing the feature to go out of spec.

The EL value of 0.742 indicates the robustness of the resist to dose fluctuations. The wider the mouth of the parabola in the graph, the higher the latitude.

The best focus being slightly negative is common for thick resist stacks, as the optimal focal point is located inside the bulk of the material rather than exactly on the surface.

S2. 18

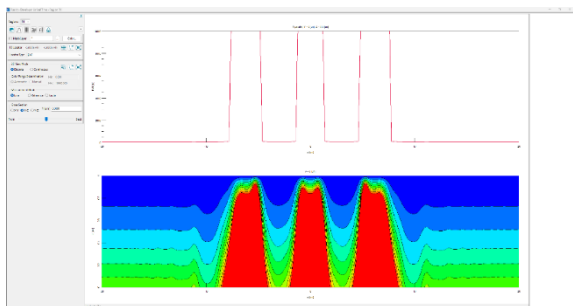


Figure 52 Discrete filled contour XZ image of the resist

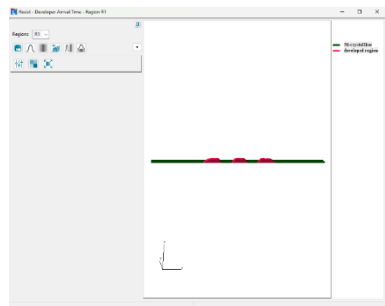


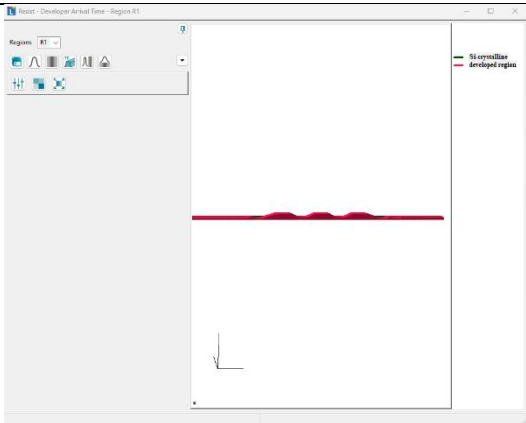
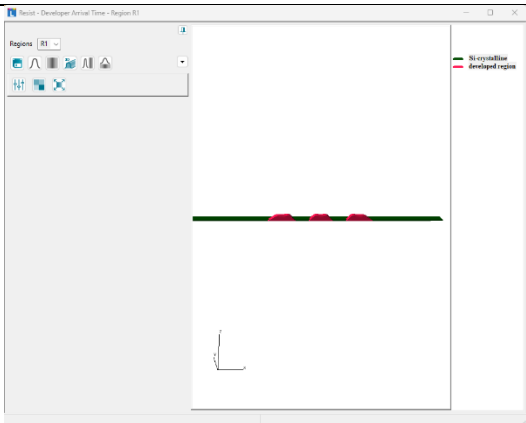
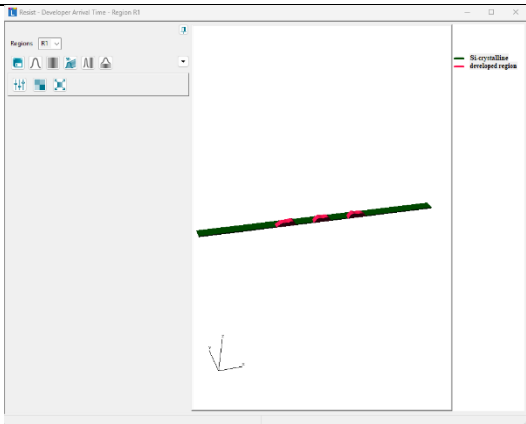
Figure 53 3D image of resist

Observation:

The three vertical red pillars represent the areas where the exposure dose was high enough to change the resist chemistry. In a positive resist, these are the regions that will be dissolved by the developer.

The red regions, which are wider at the top and narrower at the bottom, are caused by a combination of beam divergence and absorption as the light travels through the 0.5 μm thick layer.

The 3D image of the resist shows what would remain on the silicon wafer after the development step is complete. The three features are clearly separated, which shows that our choice of NA and dose was sufficient to resolve features without bridging [2].

3D image	Observation
 Figure 54 3D image of resist for dose=5	Underdeveloped: The red pillars are extremely thin and shallow. Most of the resist remains on the substrate, as the energy was insufficient to trigger enough chemical change for the developer to remove the intended regions.
 Figure 55 3D image of resist for dose=10	Partially developed: The pillars are becoming more defined and taller, but they are still rounded and do not fully reach the intended vertical height. There is still significant residual resist between the features.
 Figure 56 3D image of resist for dose=15	Optimal development: The three features are distinct, isolated, and have reached their full height. The pacing between the pillars is clean, showing that the developer has successfully reached the substrate.

Because of the above reasons, the resist is considered well developed at the dose of 15 mJ/cm².

Table 10 Exposure dose for optimal development of the resist

Height	Exposure dose (mJ/cm ²)
Top	3.02368
Center	8.56826
Bottom	14.0617

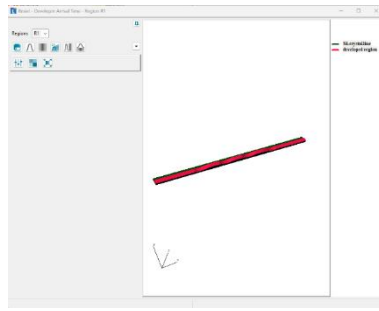


Figure 57 Before development

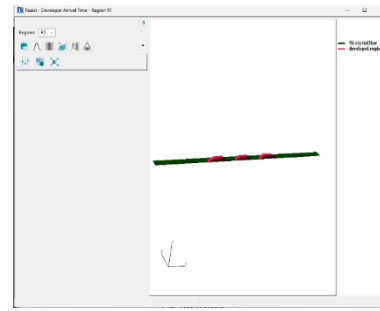


Figure 58 After development

Observation:

As the light passes through the resist, it is increasingly absorbed by the photoactive compounds. It implies that the concentration of light at the Bottom is significantly less than in the Top. Thus, in order to bring the developed size to the same size at the bottom, we need to heavily overexpose the Top of the resist.

This exposure dose of 14.0617 at the Bottom indicates that the Top of the resist is being bombarded about 4.6 times more than the amount of energy it actually requires. This may cause top-loss, where the feature becomes wider than the purpose.

In a realistic production process, we cannot adjust the dose to depth and must therefore choose one Nominal dose. Most likely, compromising the center is the most frequent. It is guaranteed that the bulk of the feature is right, but it might take more development time to guarantee that the base is cleared out [5].

S2. 21

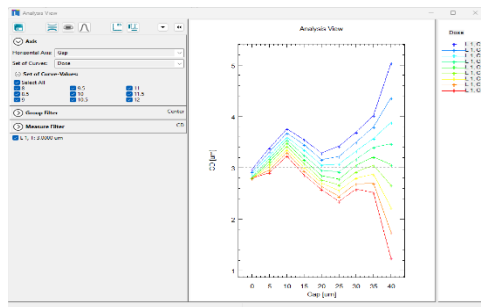


Figure 59 Dose gap matrix analysis

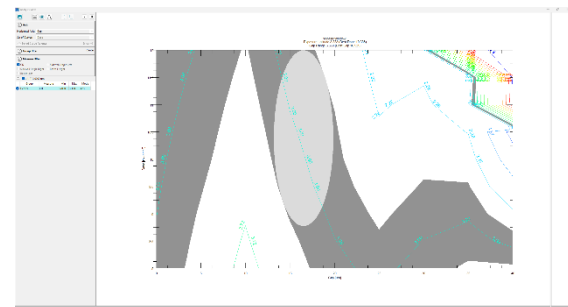


Figure 60 Exposure dose and gap range

Observation:

Exposure Latitude (EL): 3.223 Gap range: 6.663 μm

S2. 22

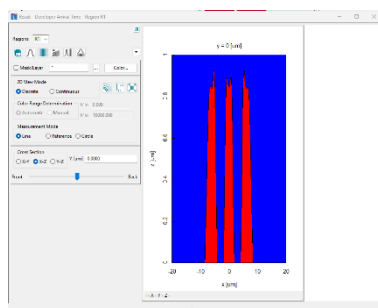


Figure 61 XZ cross-section before PEB

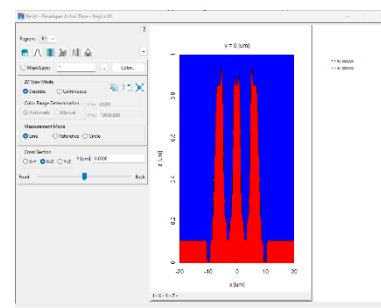


Figure 62 XZ cross-section after PEB

Observation:

PEB is a spatial filtering mechanism that behaves as a low-pass filter to eliminate the high-frequency noise, i.e. standing waves of the aerial image, but leaves the low-frequency signal, i.e. the desired mask pattern.

Before PEB: The XZ cross-section displays some strong horizontal ripples along the resist features sidewalls. They are caused by standing waves and cause periodic changes in the concentration of the Photo-Active Compound (PAC).

After PEB: The sidewalls have become significantly smoother, since we have introduced heat and triggered molecular diffusion. It permits the high and the low concentrations of the PAC to blend into each other, effectively sweeping away the vertical interference fringes [3].

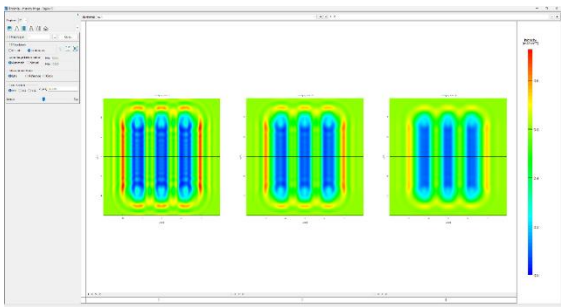


Figure 63 2D continuous intensity

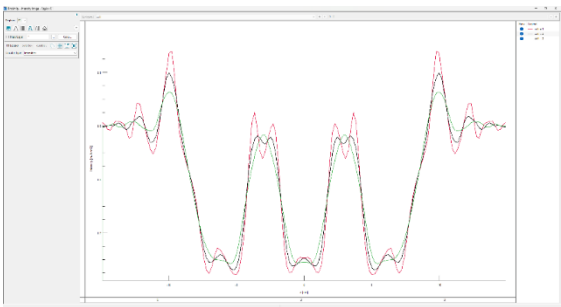


Figure 64 1D overlaid intensity

Observation:

Collimation angle refers to the degree of beam parallelism that impacts the aerial image in a lithography system.

In the 2D continuous intensity plot 0° , 2° , and 4° angles are compared, where the 0° plot indicates perfect collimation. The features are clearest, and the background interference ripples are very detailed. This is because all the rays of light are moving in one direction, which maximizes coherent interference.

As the angle increases, these small ripples begin to wash out. The edges of the three main vertical lines become more diffused. This is a form of spatial averaging [1].

Table 11 Trend from low to high collimation angle

Feature	Low collimation angle (Red curve)	High collimation angle (Green curve)
Peak intensity	Much higher and sharper	Lower and broader
Minimum intensity	Deeper (closer to zero)	Higher (shallower wells)
Contrast	High	Low
Interference ripples	High amplitude	Low (smoothed out curve)

While the perfectly collimated beam is the theoretically optimal choice, it is also sensitive to all the diffraction artefacts and interference ripple. These ripple-like oscillations in a real-life fabrication setup may lead to undesired surface roughness in the resist.

Although the smaller the collimation angle, the sharper and the better the resolution, it is sometimes helpful to make the angle a little larger in order to enhance DOF. By introducing a range of angles, the system becomes less sensitive to the exact focal plane, averaging out the focus error across the thickness of the resist [1].

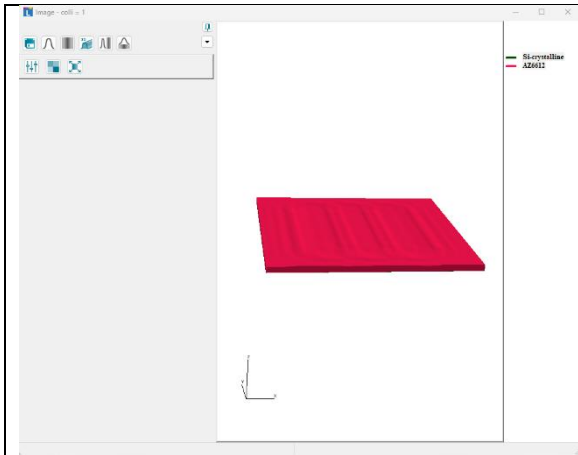


Figure 65 3D resist profile for collimation angle = 1

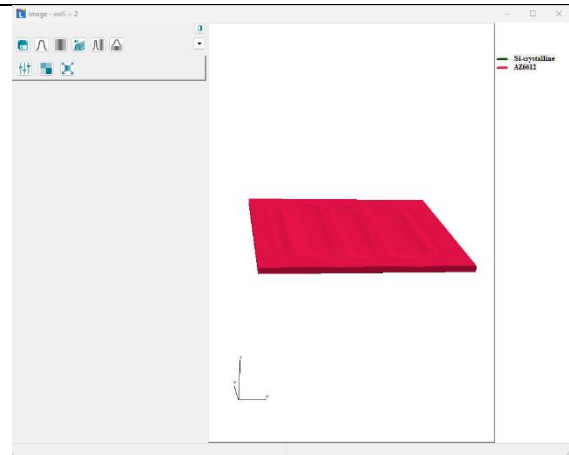


Figure 66 3D resist profile for collimation angle = 2

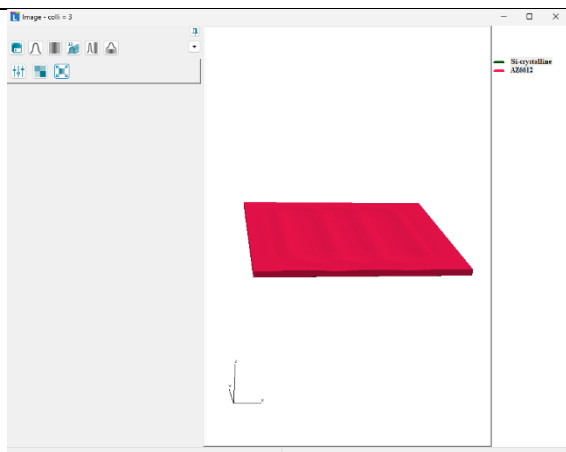


Figure 67 3D resist profile for collimation angle = 3

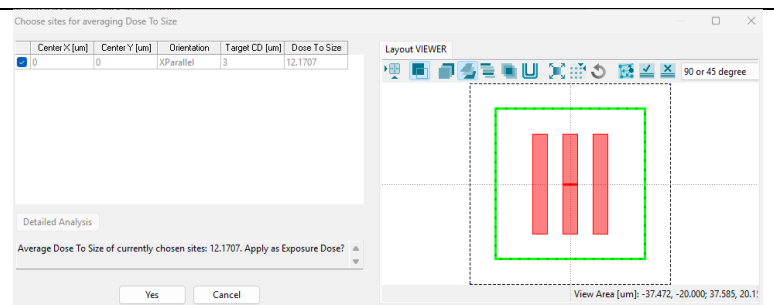


Figure 68 Optimized dose to size (bottom) for collimation angle = 2

Observation:

As the collimation angle increases, the following changes occur:

With an increase in collimation angle, more off-axis light rays are introduced. The light penetrating the masked regions would bleed into those regions, making the features wider at the top and narrower at the bottom to achieve a positive resist process.

At less collimation, the beam is more vertical and coherent. This results in a sharp edge and high contrast. The aerial image becomes less sharp at an angle of 3 °, and the top of the resist features appears smoother and blurred.

A small increment in the collimation angle, such as 1 ° or 2 °, can serve as a natural spatial filter and smooth the small interference ripples on the surface of the 3D resist, as can be seen in the transition between Figures 62 and 63.

Optimized dose (Bottom) for collimation angle = 2 is 12.1707 mJ/cm²

This particular dose is sufficient to have the formed feature at the bottom of the resist perfectly aligned with the desired CD of 3 μm. Since at this collimation angle the energy is dispersed, it is common to need a higher dose than a perfectly parallel beam of a smaller collimation angle [1][4].

S3. 25.1

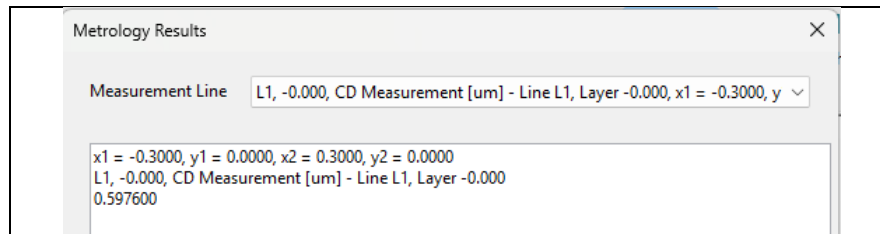


Figure 69 CD measurement for bottom metrology

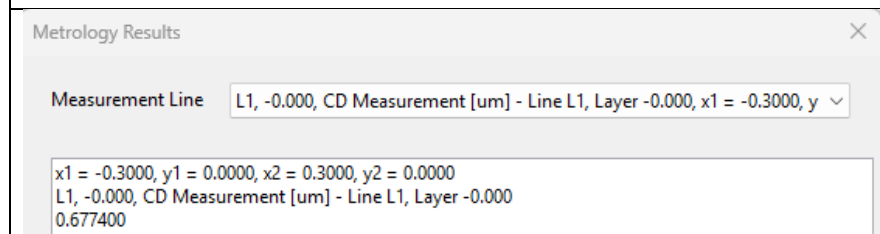


Figure 70 CD measurement for center metrology

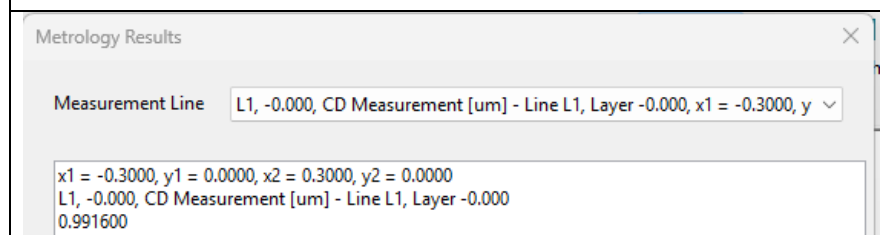


Figure 71 CD measurement for top metrology

Observation:

The CD increases as we move from the bottom to the top of the resist. The feature has a width of approximately $0.4\text{ }\mu\text{m}$ at the top compared to the bottom. This means a trapezoidal or tapered profile. This occurs in a positive resist process, since the developer has far more time to remove the top of the resist than the bottom; diffraction and absorption also contribute to it.

All three measurements are based on a common set of coordinates, but the values obtained are the actual physical width of the resist left. In the lower CD, there is a huge constriction of the opening at the base. This implies that the dose or development period of the exposure was not adequate to clear the bottom of the resist completely to align with the top size [5].

S3. 25.2

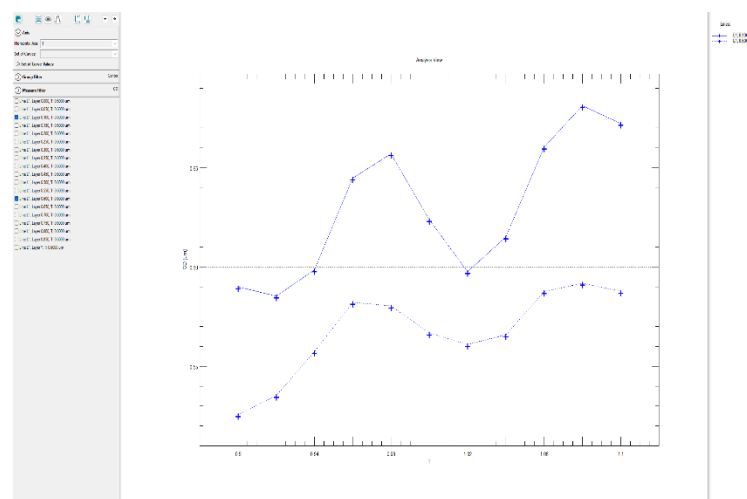


Figure 72 Results from $0.1\text{ }\mu\text{m}$ and $0.6\text{ }\mu\text{m}$

Observation:

Based on the Analysis view graph, we are looking at the CD stability as a function of the resist thickness (T) at two different vertical positions within the resist: $0.1\text{ }\mu\text{m}$ (near the top) and $0.6\text{ }\mu\text{m}$ (at the bottom).

At $0.1\text{ }\mu\text{m}$, the CD is constantly higher, generally fluctuating between $0.58\text{ }\mu\text{m}$ and $0.68\text{ }\mu\text{m}$, and at $0.6\text{ }\mu\text{m}$, the CD is constantly lower, fluctuating between $0.52\text{ }\mu\text{m}$ and $0.59\text{ }\mu\text{m}$. This confirms that the feature is significantly wider at the top than at the bottom,

confirming a positive sidewall taper. In a positive resist process, this denotes that the upper part of the resist has been better exposed or has been in contact with the developer longer [2][4][1].

In both instances, we see a periodic wave pattern as ‘T’ varies. This is the swing effect. As we change the thickness of the resist, the phase of the light reflected from the substrate changes. This causes interference with the incoming light, effectively changing the dose the resist receives, even if the source intensity remains constant.

The CD at 0.1 μm is much more sensitive to thickness changes than the CD at 0.6 μm ; the swing effect is larger for the top layer, meaning small errors in resist coating thickness will cause large variations in the final line width.

S3. 25.3

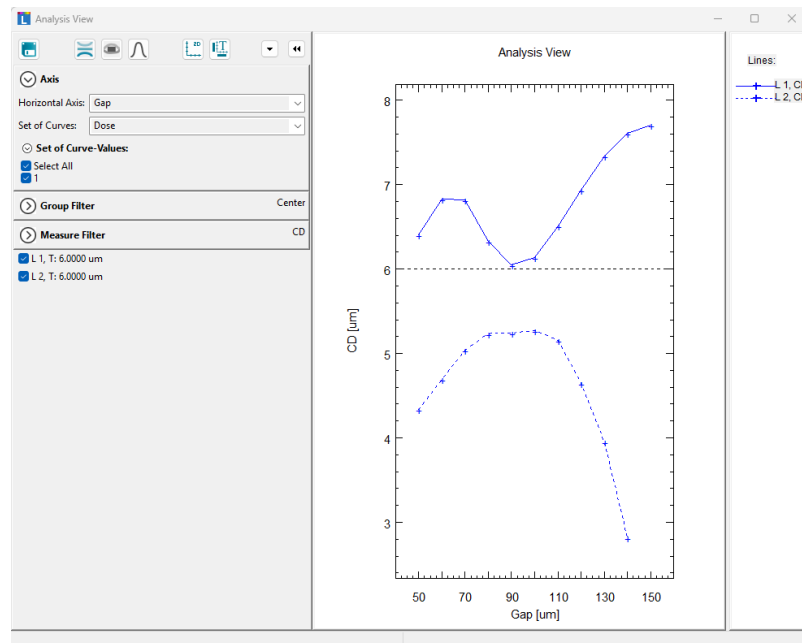


Figure 73 Analysis view results

Observation:

The most striking difference is that L1 and L2 respond to diffraction in almost exactly opposite ways:

For L1: The CD is always higher than the target line of 6 μm over almost the entire gap range, indicating that L1 is a bright object, such as a hole or a space that is expanding as the gap widens. With a gap greater than 90 μm , the CD immediately rises to almost 7.8 μm as compared to 6 μm . This means L1 is highly sensitive to the proximity gap; a slight error in the distance between the mask and the wafer will result in an astronomical error in the end part size.

For L2: A CD always lies below the target line of 6 μm , implying that L2 is probably a dark spot, such as a line of chrome, which is fading away on its edges due to diffraction. It presents a comparatively constant horizontal between 80 μm and 110 μm , but when the separation goes above 110 μm , the CD falls to 2.8 μm with a crash. It is a case of a resolution cliff in proximity printing, where the diffraction pattern has all contrast lost, and the feature is effectively lost.

Feature L2 is a much more robust design for manufacturing because it maintains a relatively stable size over a 30 μm gap range, while L1 is highly volatile [1].

Lab Assessment

LA1

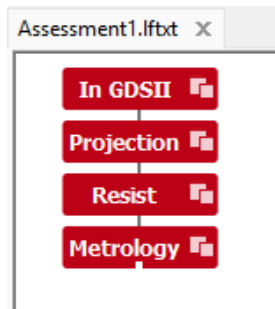


Figure 74 Process flow



Figure 75 Partially developed resist

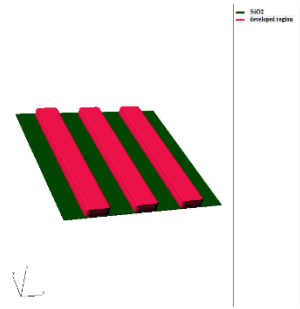


Figure 76 Fully developed resist

Process flow: The simulation was conducted using a standard projection lithography workflow.

Step 1: Import T_3micron.gds layout.

Step 2: Insert the Projection module from the lithography tools, the Resist module to load the parameters for AZ3312 resist from the database, and lastly import the Metrology module to check the grown height of the resist. Connect all of them together as shown in Figure 74.

Step 3: Run the simulation

At this time, since the optimized exposure dose is not selected, we will get a partially developed resist as shown in Figure 75.

Optimized exposure dose = 100.096 mJ/cm^2

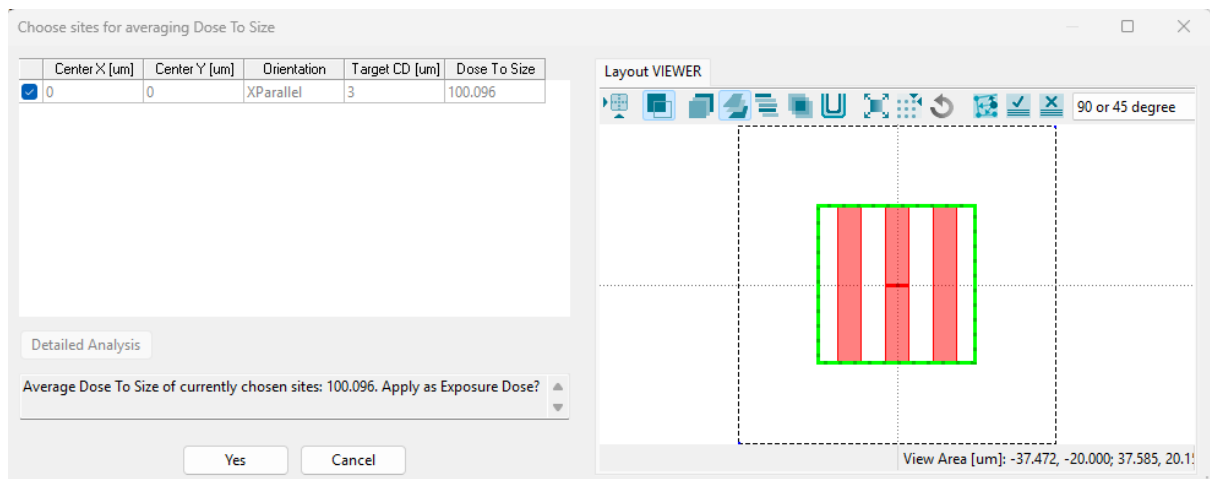


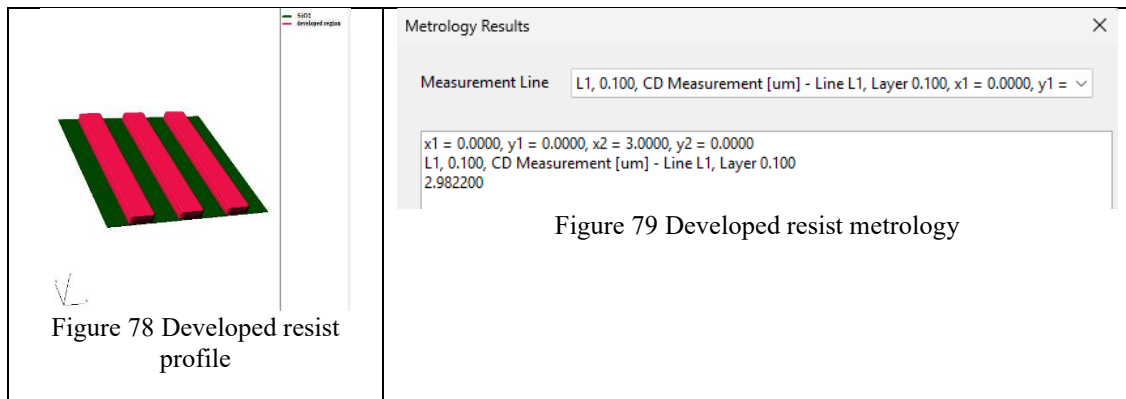
Figure 77 Optimized dose for exposure

Step 4: Double-click on the Projection module and in the metrology section define the target CD as $3 \mu\text{m}$. Now, go to the dose to size window and select the Resist (bottom). Click "ok", then we will get the optimized exposure dose as 100.096 mJ/cm^2 , as shown in Figure 77.

Step 5: Run the simulation again, this time we get the fully developed resist as shown in Figure 76. Go to the metrology result section and check the resist height at $0.1 \mu\text{m}$. It will show a height of $3 \mu\text{m}$.

For 50% more than the optimized exposure dose:

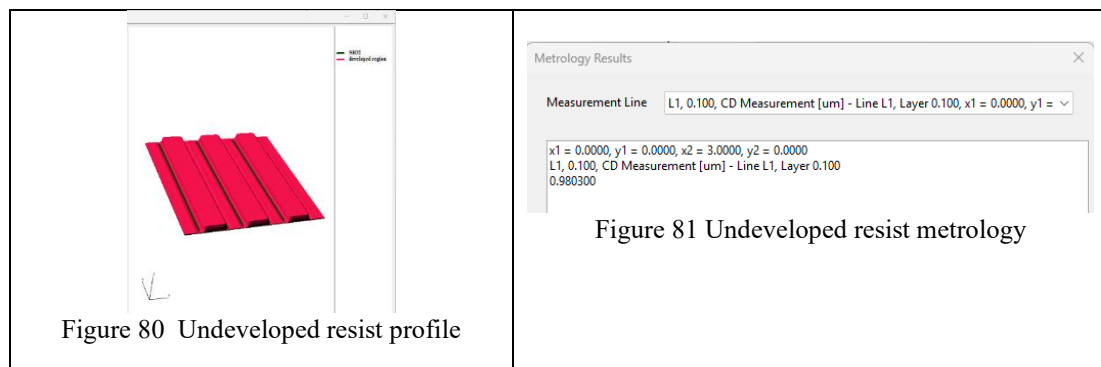
Step 6: Increase the exposure dose to 150 mJ/cm^2 . The resist remains fully developed, but the lines appear significantly thinner as shown in Figure 78.



Excessive energy causes over-exposure; the intensity threshold moves further into the masked regions, eroding the resist beyond the 3 μm design. This leads to a loss of CD as shown in Figure 79, and potentially weaker structural stability of the remaining resist lines.

For below half optimized dose:

Step 7: Decrease the exposure dose to 50 mJ/cm^2 . This represents insufficient energy to trigger enough chemical change through the full 1 μm thickness, and there is residual resist that blocks the intended pattern.



The metrology result of 0.98 μm in Figure 81 refers only to a small portion of the resist that managed to dissolve, leaving a profile that is essentially unusable for etching or deposition.

Conclusion:

At optimized dose: Optimal resist growth without scumming or thinning of the resist.

At 50% more than the optimized dose: Over-exposure and thinning of resist.

At below half the optimized dose: Under-development and residual resist leaves a profile that is unusable for anything.

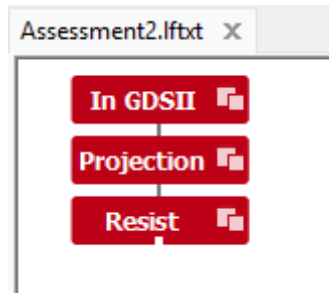


Figure 82 Before OPC process flow

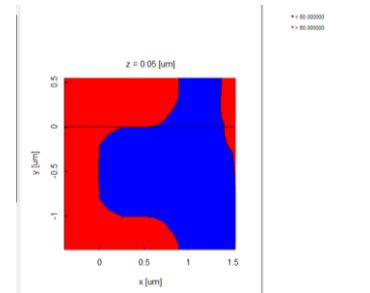


Figure 83 Artefact for projection lithography

Process flow (Before OPC):

Step 1: The FET.gds module is imported.

Step 2: The Projection and Resist modules are inserted from the lithography tools section and are connected as shown in Figure 82.

Step 3: Run the simulation. In the result section of the resist module, we find the artefact we have to correct using OPC.

Artefact replication:

The most prominent artefact is the rounding of the sharp 90° corners of the FET structure as shown in Figure 83. High-frequency spatial information needed to define a sharp corner is lost by the projection lens.

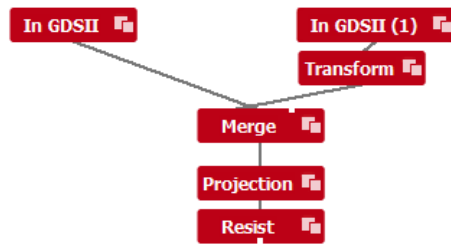


Figure 84 After OPC process flow

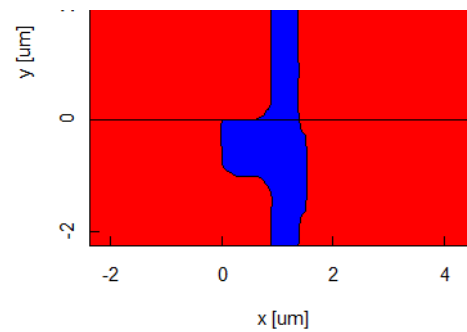


Figure 85 Resist after OPC

Process flow (After OPC):

Step 4: Import Rectangle.gds module, i.e. GDSII (1).

Step 5: Insert the Transform and merge module from the lithography tools section and connect them as shown in Figure 84.

The Rectangle.gds module contains a small correction rectangle called “Serif”. The transform module was used to precisely align this correction rectangle with the corner that suffered from rounding.

The original FET (Figure 82) layout and the new correction rectangle were combined using the “Merge” module before being sent to the Projection module.

Placement of the Serif:

We place the Serif to the outer corner of the FET module, which will compensate for the light that will inevitably be lost to diffraction, as shown in Figure 86.

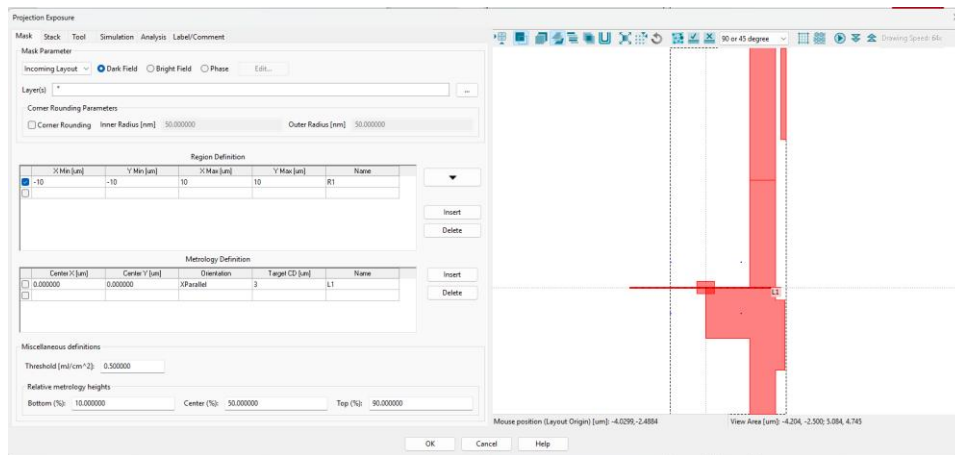


Figure 86 Representation of the applied OPC

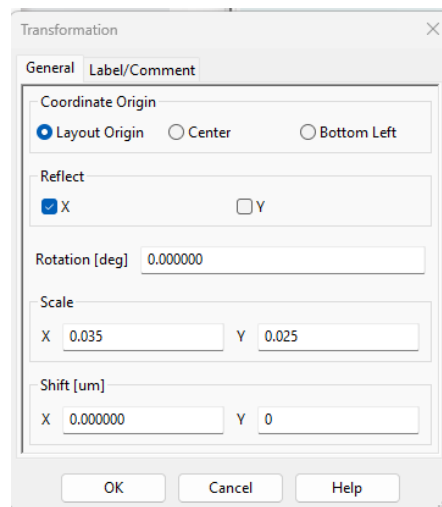


Figure 87 Changes made in the Transformation tool to find correct position to place the Serif

After OPC results:

The 2D resist image in Figure 85 shows a much squarer corner as compared to the rounded corner from Figure 83, which improves the corner fidelity.

The addition of the rectangle ensures that after development, the final resist footprint matches the intended design dimensions much more closely.

At the critical 10% resist height, the feature is now robust enough to ensure the pattern reaches the Si substrate with the correct geometry.

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